BFRNet: Bidimensional Feature Representation Network for Remote Sensing Images Classification

Wei Wang, Ting Hu, Xin Wang, and Ji Li

***Abstract*— In recent years, convolutional neural network (CNN) and transformer, as mainstream classification methods, have made good progress in improving the classification performance of remote sensing (RS) images. Furthermore, the CNN-transformer hybrid architecture has shown a greater potential for enabling models to obtain local information and global dependency relationships. In general, numerous researches improve self-attention block of vision transformer (ViT) in the spatial dimension. Nevertheless, spatial self-attention mostly achieves a single spatial feature extraction, which cannot meet the requirement of accurate recognition of high-resolution RS images. In this work, a method for representing spatial and channel dimensions of RS images is proposed, which not only extracts global–local spatial features but also pays special attention to incorporate channel information. Specifically, bidimensional local window self-attention (BLWS) and pyramid pool self-attention are conducted to extract local–global features. Subsequently, a linear attention module will fuse local–global information in the channel dimension when computing multihead self-attention (MHSA). A bidimensional gating unit (BGU) is used to replace the traditional multilayer perceptron (MLP) of the feedforward network (FFN). The above improvements result in a bidimensional feature representation (BFR) block, and BFR network (BFRNet) is designed based on BFR blocks. BFRNet consists of four stages, and each stage repeatedly stacks BFR blocks with different layers. Experiments show that the classification accuracy of BFRNet is significantly better than the existing methods of CNN, ViTs, and CNN-transformer networks. On dataset RSSCN7, BFRNet achieves a classification accuracy of 98.75% with only 1.9G floating point operations (FLOPs), which is 8.21% higher than ViT, 3.21% higher than Resnet50, and 2.68% higher than CoAtNet, respectively.**

***Index Terms*— Channel attention, convolutional neural network (CNN), image classification, remote sensing (RS), spatial attention, transformer.**

I. Introduction

THE rapid development of remote sensing (RS) technology has provided a rich source of data for high-quality RS images. The effective way of processing RS images is an important topic in this research area [1]. RS image classification is a process of analyzing the scene content of RS images and achieving semantic label assignment, which is beneficial for applications, such as traffic analysis, land use, and land cover [2]. RS image classification can be performed by traditional convolutional neural network (CNN) classification methods. Wu et al. [3] proposed a classification framework by embedding the cross-channel reconstruction mechanism as a plug and play module into CNN, which can obtain richer feature representations and reconstructs different features across channels. In recent years, vision transformer (ViT) has received massive attention and achieved great success in various vision tasks, and its success lies in the application of self-attention mechanism, which obtains contextual information by capturing the relationship between global tokens and local tokens [4]. ViT divided the feature maps into image sequences for feature extraction [5]. After that, Swin Transformer divided the fixed-size token sequences into separate blocks of different sizes according to the hierarchy to obtain more information with sliding window attention [6]. However, the quadratically increasing complexity of the self-attention of the ViT series model limits its faster application to high-resolution images.

Currently, both CNN and transformer are the mainstream RS image classification methods, and CNN has been showing excellent performance and potential. In general, CNNs have a faster inference speed than the ViT-based methods and thus remain the dominant approach for vision tasks. CNN is a data-driven approach that learns more representative and essential features through an end-to-end hierarchical approach with properties, e.g., parameter sharing and powerful local information extraction [7]. A recent study proposed a CNN framework that not only enabled pixel-level classification, but it was also applicable to other CNNs for spatial information modeling [8]. Resnet, as a classical CNN, improved the degradation problem of deep networks in terms of residual learning to achieve milestone-level improvements [9]. Later work was devoted to smaller and lighter models, such as mobileNet [10]. A lightweight neural network focused on mobile device deployment, in which depthwise separable convolution was denoted as deepwise convolution and pointwise convolution, respectively. Tan and Le [11] proposed an efficient model scaling method to scale up CNNs in a structured manner, and the network model achieved higher accuracy. Le et al. [12] proposed a new compressed network that compressed the weights of the standard fully connected layer and the kernel size of the convolution to

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reduce parameters. The new fully connected layer structure takes full advantage of the intrinsic correlation between image information to enhance its feature extraction and classification capabilities. The recently proposed ConvNeXt by FAIR relied on the convolutional structure and achieved comparable top-1 accuracy of the popular transformer-based neural networks [13].

Despite the great success of CNN-based models, the implementation in RS images still suffers from some serious problems. For instance, CNNs use convolution to extract high-level features, but the convolution is limited by the local receptive field, which makes it difficult to capture long-distance dependencies information. However, ViT has powerful global information extraction capabilities, although it increases the complexity of the model. In order to enhance the lightweight property of ViT and improve their performance in vision tasks, recent work has focused on developing various ViTs in a hybrid or heterogeneous manner and drew on the advantages of local feature extraction from CNNs, such as integrating convolution into transformer block as plug and play modules. For example, ResT used convolution to reduce the spatial dimension of keys (K) and values (V) in self-attention [14]. Wang et al. [15] constructed multiscale attention blocks by convolution and then fused them into transformer to form a visual method for local–global feature extraction. MTCNet introduced a bidimensional attention block composed of convolution into transformer to obtain the multiscale features of RS images [16]. The recursive gated convolutional block proposed in HorNet can be used as a plug and play module to improve various ViT-based and convolution-based models [17]. In order to extract low-level features and capture multiscale attention for high-level features, CoAtNet employed convolution for local self-attention [18]. Based on a hybrid CNN-transformer architecture, Sun et al. [19] used 3-D convolutional layers and 2-D convolutional layers to extract shallow features and tokenize features in transformer. In addition, some approaches, such as GCVit and MaxViT, improved CNNs by inserting self-attention into the existing CNN blocks [20], [21]. Chen et al. [22] integrated transformer into MobileNet [10]. In MobileViT, transformer is integrated into MobileNetV2 to obtain a global representation of the upper level [23], [24].

Generally, the local information of RS images contain low-level features and rich spatial information, which can capture more feature of small targets, but the receptive field of pixels is small. Meanwhile, global information contains high-level features and rich semantic information, which can capture the feature of large and medium targets. At the same time, as the network deepens, the receptive field becomes larger. Therefore, global features and local features complement each other. Many works are carried out based on global–local feature extraction. For example, Liu et al. [25] proposed an innovative HSI-TNT framework, in which local information between pixels was extracted by internal T-blocks and global information was extracted by external T-blocks. By using the multilevel semantics of complex scenes, Zhu et al. [26] proposed deep-local–global feature fusion (DLGFF) framework for high-resolution scene classification. He et al. [27] proposed DGLAnet, which used bidirectional relationships to capture spatial and spectral information to obtain global concerns, while obtained local spatial and spectral features through pooling layers.

Global contextual information is the key to RS image classification, but the localization of convolution is also essential [28]. Therefore, bidimensional feature representation network (BFRNet) proposes a hybrid neural network model based on CNN-transformer architecture, which implements the fusion of local features with global features. In this work, bidimensional local window self-attention (BLWS) is used to extract local features with channel information. Local window attention uses nonoverlapping windows to achieve self-attention inside the window and takes into account channel dimension feature extraction. So, BFRNet changes the previous way of extracting single spatial dimension information using only spatial self-attention, which obtains richer information. BLWS tends to extract local feature, and a pyramid pool self-attention is used to extract global contextual information.

Throughout the modeling process, BFRNet focuses on capturing channel information while extracting spatial information. Thus, a linear attention module is used for global–local feature interaction attention. Overall, the contributions of this article are as follows.

1) In order to reduce the computational complexity of ViT and improve the limited receptive field of CNN, an innovative and efficient ViT block, namely, the BFR block, is proposed, and an advanced CNN-transformer hybrid neural network (BFRNet) is constructed by stacking BFR blocks.
2) BFRNet aims to design a model with BFR and the ability to capture both global and local information. Therefore, a joint attention module called BFR-multihead self-attention (MHSA) is designed to improve the MHSA of ViT by combining local–global information from the perspective of spatial information and channel information fusion.
3) The proposed BFRNet is a powerful neural network, and extensive experiments show that BFRNet has significant efficiency, as shown in Fig. 1, and BFRNet achieves 98.75% accuracy on RSSCN7 dataset, which is significantly better than the existing visual neural networks.

The rest of this article is structured as follows. Section II is a related work that describes the current state of development in image classification and the background of the methods. Section III describes the structure of BFRNet, including the BLWS, the global pyramid pool self-attention, the linear attention module, and the bidimensional gating unit (BGU). Section IV presents the experiments, including experimental preparation, experimental results analysis, and ablation experiments. Section V is the conclusion.

II. Related Work

A. Efficient ViTs

Transformer was originally proposed in the field of natural language processing (NLP), which relied on MHSA

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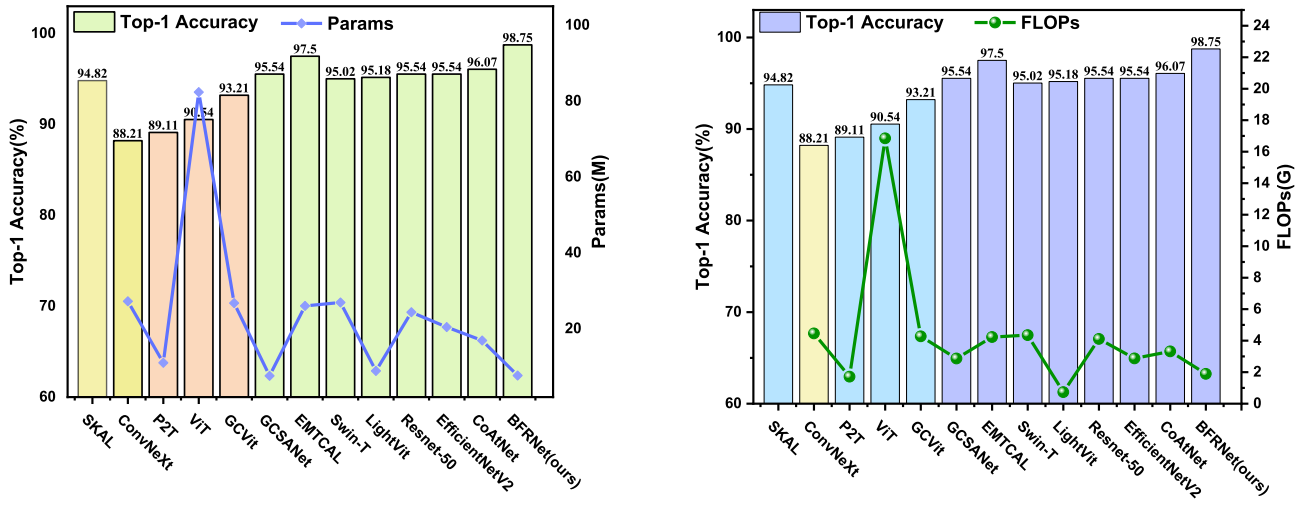


Fig. 1. Left: parameters of contrastive networks and BFRNet. Right: GFLOPs of contrastive networks and BFRNet.

to demonstrate excellent global modeling capabilities from token to token. Transformer has attracted a lot of attention from researchers and has been introduced to the field of computer vision since global relations are crucial for vision tasks. Dosovitskiy et al. [5] proposed ViT, which contributes greatly to vision tasks by dividing images into small patches and executing self-attention between these patches. DeiT reduced the training cost of ViT by knowledge distillation [29]. T2T-ViT designed a new token-to-token process to overlap segmented images, which enhanced the local relationships between tokens instead of reducing their length [4]. Wu et al. [30] introduced deep convolution to generate Q, K, and V in generating MHSA. In order to reduce the number of tokens used in computing MHSA, MViT employed a single pooling operation [31]. MViT actually performed the modeling of token–region relationships instead of the traditional token-to-token modeling. This operation also helps to reduce the length of the token sequence. Heo et al. [32] added a pooling layer based on ViT and demonstrated its effectiveness experimentally. BFRNet is followed this idea by using pyramid pool to reduce the sequence length and the model training cost. Based on the CNN-transformer framework, SpectralFormer fused the "soft" residuals through adaptive learning to transfer features from shallow to deep layers in a memory-like manner [33]. Numerous studies have shown that operations, such as pooling, training strategies, improve self-attention, and framework binding, can be flexibly coordinated with ViT. These strategies have improved the performance of ViT and pushed it forward.

B. Window-Based ViTs

ViTs are successful in image classification but still face challenges when used for high-resolution image recognition tasks. Some recent works, e.g., Twins, Swin Transformer, Pyramid ViT, and Pvt v2, use a CNN-like multistage hierarchical structure to construct a more efficient ViT [34], [35], [36]. Among these networks, the local window-based approach uses a local window sliding mechanism to obtain a larger receptive field and reduce the complexity [5], which gets the global information ignored by the nonoverlapping local windows and expands the receptive field in this way. Huang et al. [37] believed that spatial shuffling was an effective way to establish connections between windows and recognized that the cross-window connections were key to improve the image representation. Yu et al. [38] showed that the shift window partitioning was not essential and effective cross-window communication can be achieved by a deep convolution alone. Therefore, the complex window shift partitioning is discarded and a nonoverlapping local window partitioning method is used in BFRNet. Nonoverlapping local window is self-attention inside each window, so that the computational complexity is linearly related to the image size.

More importantly, considering the small receptive field of local windows and the weak global dependencies, new approaches by introducing global interactions into local windows have been proposed in many works. MSG-Transformer fixed learnable message tokens on each local window and employed channel shuffling to exchange information between these tokens [39]. Yang et al. [40] performed local window attention using keys (K) and values (V) of different strides to achieve the effect of aggregating information captured from different sensory fields. The interaction of fine-grained features extracted by the local window attention with coarse grained global features enhances the image representation effectively.

C. Transposed Self-Attention

The most expensive computational cost of SA in traditional ViT comes from the self-attention [41], whose operation interacts the query (Q) with the key (K) in dot-product along the spatial dimension, i.e., $Q \odot K^{\mathrm{T}}$. It results in a quadratic growth in computational complexity as the resolution of the input feature map increases, i.e., for an $H \times W$ image, the operation is $(HW \times C) \odot (C \times HW)$ with complexity $O(H^2W^2)$. The traditional SA operation is avoided in EdgeNeXt [42]. Instead of using $Q^{\mathrm{T}} \odot K$, its operation implied that in channel dimension query (Q) interacted with

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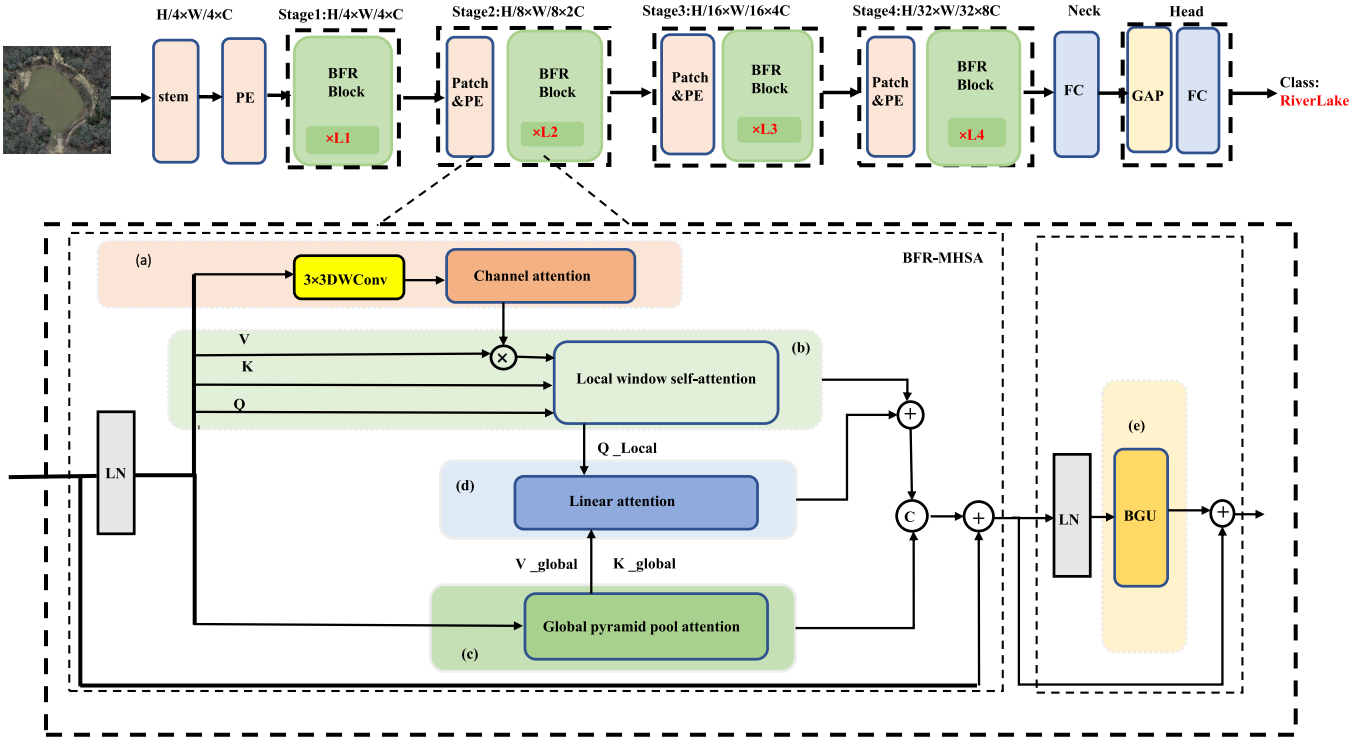


Fig. 2. Overall architecture: modules (a)–(e) make up the BFR block, and BFRNet consists of four stages, each of which has different layers of BFR block.

key (K), i.e., $(C \times HW) \odot (HW \times C)$. So, $C \times C$ softmax scaled attention scoring matrix with linear complexity was generated. This transposed self-attention mechanism has linear complexity, which is also used in Restormer [43]. It is the key to communicate across channels rather than across spatial dimension. Transposed self-attention can extract different information from another dimension as a complement to the spatial information.

III. Methodology

In this section, we first describe the overall architecture of BFRNet and then provide a detailed description of the core design. Detailed description includes the bidimensional local window attention (BWLS), pyramid pool self-attention, linear self-attention, and BGUs.

A. Overall Architecture

Following the general approach of previous work, we rethink two key components of ViT: i.e., the self-attention and the feedforward network (FFN), to form the novel feature extraction method proposed in this work. For the self-attention part, BFR–MHSA is used to extract richer information with global–local feature fusion, i.e., local features are extracted by BLWS, which contains both spatial and channel information, and global features are extracted by pyramid pool self-attention. Also, we note that spatial self-attention extracts image information only in the spatial dimension, so channel dimension self-attention (linear attention) is added to interact with BLWS and pyramid pool self-attention module. In addition, considering the spatial limitation of local window self-attention for image feature extraction, channel attention is added to enhance the capability of feature extraction. The overall architecture of BFRNet consists of four stages, in which BFR blocks of different layers are superimposed to extract multiscale features, as shown in Fig. 2. First, the input feature map of size $H \times W \times C$ is passed through the stem to obtain the feature map of size $H/4 \times W/4 \times C$. Then, the output is passed to the BFR block in stage 1 for feature extraction to output the feature map $H/8 \times W/8 \times 2C$. The output is passed to the next stage in turn, and finally, the feature map of size $H/32 \times W/32 \times 8C$ is output after stage 4. To obtain more feature output, the output of stage 4 is passed into the neck. After that, the output of the neck is passed through the head to get the final classification label. The specific layers of BFR blocks in stages 1–4 are set as [2, 2, 6, 2], and the process of BFR block can be expressed as follows:

$$Z = \mathrm{LN}(X) \tag{1}$$

$$Z' = Z_{\mathrm{local}} + Z_{\mathrm{linear}} \tag{2}$$

$$M = \mathrm{concat}(Z', Z_{\mathrm{global}}) \tag{3}$$

$$y = \mathrm{BGU}(\mathrm{LN}(M)) + M \tag{4}$$

where Z_{local} is obtained from (6), Z_{global} is obtained from (7), and Z_{linear} is obtained from (8). Equations (1)–(3) indicate the calculation process of BFR–MHSA. Equation (4) indicates the output result of BFR block. LN is the layer normalization, and concat operation splices the feature maps output from the pyramid pool self-attention module (c) with the feature maps output from the summing of modules (b) and (d) along the channel dimension. BGU is the BGU, which is shown in Fig. 2(e).

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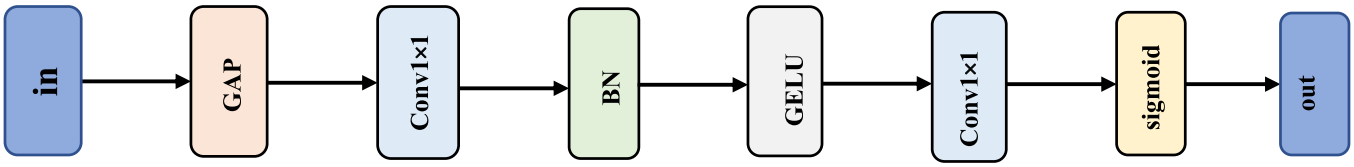


Fig. 3. Channel attention in Fig. 2(a), which consists of global average pooling (GAP), Conv 1 × 1, BN, and activation function GELU. The input feature map passes through a 3 × 3 depthwise convolution and then input to this channel attention.

B. BLWS

Despite the great performance of ViT in vision tasks, the computational cost grows quadratically with image size in high-resolution tasks of RS images. However, the local window-based approach divides the feature map into several nonoverlapping windows. Then, self-attention is performed within each local window, which makes the computational complexity linear with the image size while improving the computational efficiency. In this work, a local window attention mechanism is used as a local feature extraction method. Given an input feature map $X \in R^{H\times W\times C}$, unlike previous work where attention is paid to a patch of size $H \times W$, self-attention is paid to a locally cropped image with a nonoverlapping window shape of $(H/S \times W/S, S \times S,$ and $C)$, where S is the window size and is set to seven based on Swin Transformer. Self-attention is then applied within each local window, which is equivalent to local window attention in Twins [34].

However, the self-attention of the local window is an improvement on the self-attention of the ViT, which essentially extracts spatial information from the image, so we add channel attention to capture richer image information. Among the channel attention initially proposed, SENet has a great influence [44]. By Zhang et al. [45] used the squeeze-and-excitation channel attention block to enhance the features. Subsequent studies obtained better performance after SENet [46], [47], [48]. Since the use of fully connected layers for the squeezing and excitation modules in SE block led to a great increase in computational cost, it was replaced by 1 × 1 convolution in recent studies. This improved channel attention mechanism is used on BFRNet, as shown in Fig. 3. The channel attention consists of global average pooling (GAP), batch normalization (BN), and activation function Gaussian error linear unit (GELU) [49], and the output is Z_{channel}. The input feature map $X \in R^{H\times W\times C}$ passes through a 3 × 3 depthwise convolution and then is inputted to the channel attention, as shown in Fig. 2(a). Inspired by Chen et al. [50], we combine this attention mechanism with the local window attention mechanism to extract local information from the spatial dimension and the channel dimension. Specifically, the input feature map $X \in R^{H\times W\times C}$ is subjected to a depthwise convolution to extract features, after module (a), and then multiplies with the value (V) of the window self-attention, as shown in Fig. 2(b). The specific details of the BLWS are shown in Fig. 4.

The output of BLWS module is as follows:

$$Z_{\text{dw-channel}} = Z_{\text{channel}}(\text{DWConv}(Z)) \tag{5}$$

$$Z_{\text{local}} = \text{soft}\max\left(\frac{Q_{\text{local}}K_{\text{local}}^{T}}{\sqrt{dk_{\text{local}}}}\right)(V_{\text{local}} \otimes Z_{\text{dw-channel}}) \tag{6}$$

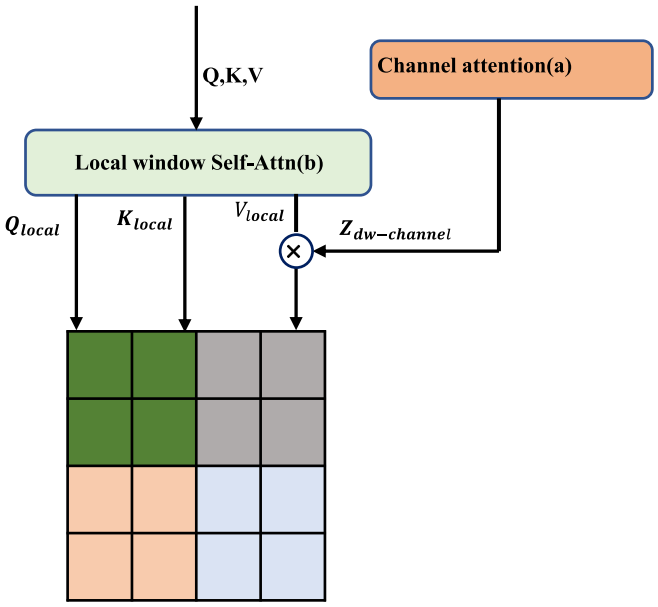


Fig. 4. Bidimensional local window self-attention (BLWS) in Fig. 2(b). BLWS consists of local window self-attention and channel attention, where channel attention is shown in Fig. 3. × denotes the matrix multiplication in the window self-attention.

where Q_{local}, K_{local}, and V_{local} are the query tensor, key tensor, and value tensor of the BLWS, respectively, dK_{local} is the channel dimension of the key tensor of the BLWS. DWConv is the 3 × 3 depthwise convolution. Z_{channel} is the output of the channel attention described in Fig. 3, and $Z_{\text{dw-channel}}$ is the output of the depthwise convolution and channel attention module (a). Equation (6) indicates the output after the combination of modules (a) and (b).

C. Pyramid Pool Self-Attention

However, BLWS tends to have weak global long-distance dependence. Therefore, global pyramid pool self-attention module is used for modeling global dependencies. Pyramid pool is a long-established and widely recognized feature extraction technique, which is important to turn a pooling into a pooling with multiple scales to avoid information loss due to cropping and warping operations. Several works are carried out on this basis, e.g., He et al. [51] introduced spatial pyramid pool into CNNs for image classification and target detection. According to P2T, pooling layers with different pooling ratios were introduced into ViTs with the aim of extracting richer multiscale features [52]. In this work, pooling with different ratios is used to calculate MHSA in each stage of BFRNet. After different scaled pooling layers, the token sequence becomes shorter, which reduces the computational cost, and the number of groups of different pooling ratios

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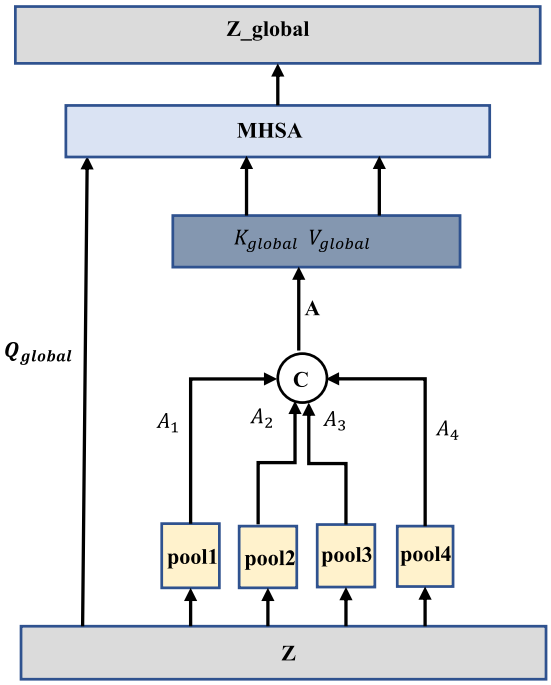


Fig. 5. Global pyramid pool self-attention is shown in Fig. 2(c). First, the input feature map is flattened into a 2-D shape, and then, four average pooling layers of different scales are applied to the reshaped 2-D image.

corresponds to the number of stacked layers of BFR blocks. In this way, BFRNet obtains shallow coarse-grained features in the early stage and deep detailed features in the middle and final stages to identify RS images accurately, as shown in Fig. 2(d). First, the input feature map is flattened into a 2-D shape, and then, four average pooling layers of different scales are applied to the reshaped 2-D image. Four stages in BFRNet correspond to four sets of pooling scales, poolratio1 [12, 16, 20, 24], poolratio2 [6, 8, 10, 12], poolratio3 [3, 4, 5, 6], and poolratio4 [1, 2, 3, 4], respectively. A set of pooling ratios are obtained after four average pooling to get A1, A2, A3, and A4, respectively, and they are spliced along the channel and then normalized by layer to get A. A contains global contextual information and can be used as a strong substitute for the feature map Z, as shown in Fig. 5, where Z is the output feature map after layer normalization of the input feature map X. The query tensor, key tensor, and value tensor in MHSA are Q_{global}, K_{global}, and V_{global}, respectively. The output result of module (c) is as follows:

$$Z_{\text{global}} = \text{soft}\max\left(\frac{Q_{\text{global}} K_{\text{global}}^{T}}{\sqrt{dK_{\text{global}}}}\right) V_{\text{global}} \tag{7}$$

where K_{global}, V_{global}, and Q_{global} are the key tensor, value tensor, and query tensor of the pyramid pool self-attention module, respectively. dK_{global} is the number of channel dimension of the key tensor in pyramid pool self-attention module.

D. Linear Attention

The computational complexity of transposed self-attention grows linearly with increasing image resolution, and unlike previous spatial self-attention, transposed self-attention is essentially channel self-attention. We follow the idea of transposed self-attention and design a new global–local feature fusion module called linear attention. The linear attention is a variation on the traditional MHSA and used to achieve the shift of the attention dimension of multiheaded self-attention from spatial to channel. The main purpose fuses local features that captured by BLWS with global features that extracted by pyramid pool self-attention, as shown in Fig. 2(c). Meanwhile, considering that the local and global branches obtain spatial information, the use of linear attention here can extract the ignored features from the channel dimension. This is done by calculating the feature map query tensor Q_{local}^{T} corresponding to the output feature map by BWLS, and the feature map key tensor K_{global} and value tensor V_{global} corresponding to the output feature map by pyramid pool self-attention are input to linear attention to calculate its MHSA, i.e., the dot-product of Q_{local}^{T} and K_{global}, as shown in Fig. 6. The output result of module (d) is expressed as follows:

$$Z_{\text{linear}} = \text{soft}\max\left(\frac{Q_{\text{local}}^{T} K_{\text{global}}}{\sqrt{dK_{\text{global}}}}\right) V_{\text{global}} \tag{8}$$

where K_{global} and V_{global} are the key tensor and value tensor of the pyramid pool self-attention module, respectively. Q_{local}^{T} is the query tensor of the feature mapping of the input Z. dK_{global} is the number of channel dimension of the key tensor of the pyramid pool self-attention module (c).

E. BGU

In transformer, the FFN is used for spatial transformation of the MHSA output. FFN introduces nonlinearity through the activation function (GELU) and increases the expressiveness of the model. In previous work, the FFN consists of multilayer perceptron (MLP) and LN, and the initial MLP is too bulky and not friendly especially for high-resolution images. Recently, Wang et al. [53] mentioned the concept of gating, which used two branch convolutions to form a gating unit with a simple structure and low computational cost. BFRNet adopts this structure, as shown in Fig. 2(e). It stipulates that the first branch uses 1×1 convolution and 3×3 depthwise convolution for capturing spatial feature, and the second branch uses only one 1×1 convolution to upward dimension as a way to obtain the channel information. After the output of the two branches, 1×1 convolution is used to downward dimension and the output is summed with the BFR–MHSA input as a feature complement. The specific process is shown in Fig. 7.

IV. Experiments

A. Experimental Preparation

The experimental data used in this article are RS image datasets: the RSSCN7 and the AID datasets. The RSSCN7 dataset contains 2800 images in seven classes, each of which contains 400 images. The image resolution is 400×400, and the scene classes are river, forests, farmlands, parking lots, and so on (the partial image examples are shown in Fig. 8).

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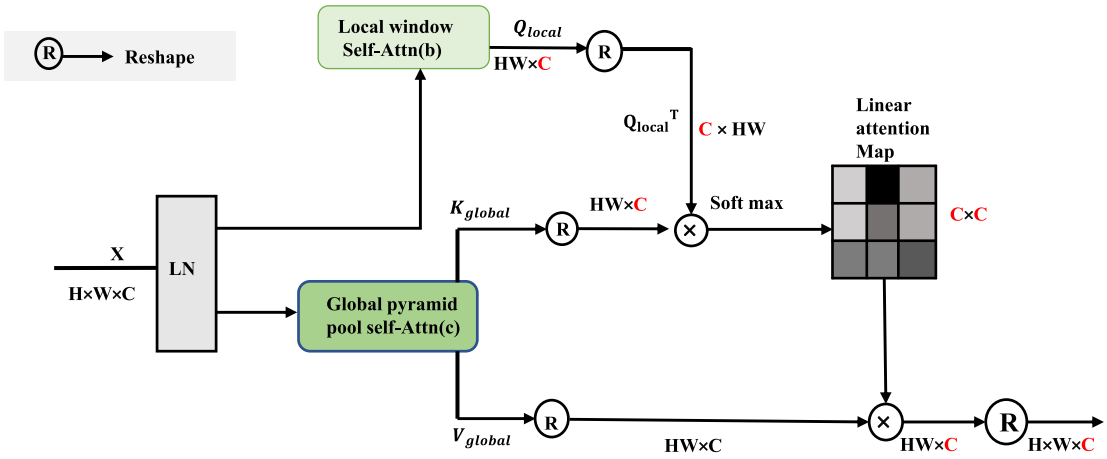


Fig. 6. Linear attention is shown in Fig. 2(d). It is done by calculating the feature map query tensor corresponding to the output feature map by BWLS, and the Q_{local}^T feature map key tensor K_{global} and value tensor V_{global} corresponding to the output feature map by pyramid pooling self-attention are input to linear attention to calculate its MHSA.

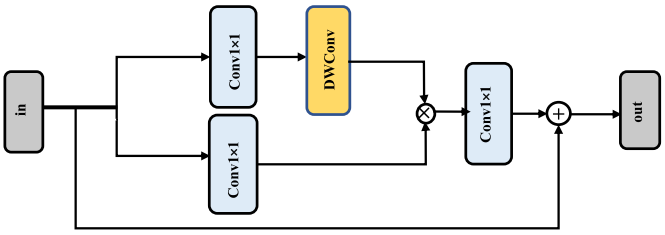


Fig. 7. Bidimensional gating unit is shown in Fig. 2(e).

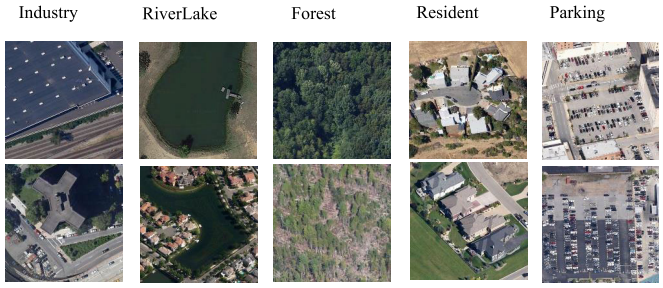


Fig. 8. Examples from the RSSCN7 dataset.

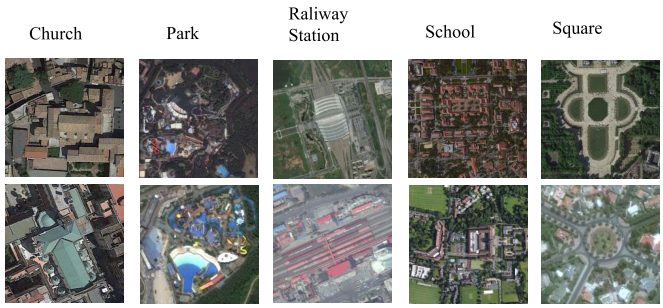


Fig. 9. Examples from the AID dataset.

The AID dataset contains 10 000 images in 30 classes, and the image resolution is 600 × 600. The scene classes include parking, school, industry, resident, forest, and so on (the partial image examples are shown in Fig. 9). Both datasets are divided according to the ratio of 2:8, i.e., 80% of the images are randomly selected as the training set and 20% of the images are used as the validation set. Both datasets are cropped to 224 × 224 size as the input feature maps of the network. We conducted two sets of comparative experiments on the RSSCN7 and AID datasets to compare with the ten networks proposed in previous work, and the ablation experiments were conducted on the RSSCN7 dataset.

Pytorch1.10.0 and Python3.9.12. are used in the experiments. The CPU is Intel[1] Xeon[1] Silver 4314, the GPU is A10, and the video memory size is 80 GB. The learning rate is set to 0.0001, the epochs are set to 400, and the batch size is set to 16.

B. Analysis of Experimental Results

In order to verify the effectiveness of BFRNet, it was compared with Resnet50, ConvNeXt, CoAtNet, ViT, and other classification methods, respectively. The experimental results on the RSSCN7 and AID datasets are shown in Table I. BFRNet achieves more excellent results compared with other methods and achieves higher classification accuracy on both RSSCN7 and AID datasets, reaching 98.75% and 95.20%, respectively. To further validate the effectiveness of BFRNet, we compare it with the latest RS image classification methods: structured key rear localization (SKAL) [57], GCSANet [58], and EMTCL [59]. The classification accuracy of BFRNet is 3.93% higher than SKAL in RSSCN7 dataset and still 1.7% higher compared to EMTCAL. SKAL designed a weakly supervised SKAL strategy and used it for AlexNet, GoogleNet, and ResNet18, which are three widely used CNN models. The number of parameters and floating point operations (FLOPs) of SKAL are not given here because there is no uniform standard for calculating the number of parameters and FLOPs for different CNN models.

Although ViT has better global modeling capabilities, the classification accuracy of ViT is slightly lower on both datasets due to its weaker ability to capture detailed information. The classification accuracy of Swin Transformer is about 3% lower than BFRNet on both datasets. Swin Transformer is a pure transformer structure and uses a local window for feature

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TABLE I
EXPERIMENTAL RESULTS OF BFRNET COMPARED WITH OTHER METHODS ON RSSCN7 AND AID DATASETS

Model	RSSCN7 Top-1(%)	AID Top-1(%)	Params (M)	FLOPs (G)
Resnet50 [9]	95.54	93.35	24.37	4.11
EfficientNetV2[56]	95.54	93.85	20.46	2.87
ConvNeXt [13]	88.21	86.90	27.24	4.46
ViT [5]	90.54	80.20	82.38	16.85
Swin-T [6]	95.02	90.40	26.93	4.35
ResT [14]	94.29	92.15	13.03	2.00
P2T [49]	89.11	90.10	11.04	1.71
LightVit [55]	95.18	91.05	8.91	0.73
GCVit [20]	93.21	90.95	26.76	4.28
CoAtNet [18]	96.07	93.65	16.92	3.32
SKAL [57]	94.82	97.05	—	—
GCSANet [58]	95.54	94.90	7.61	2.87
EMTCAL [59]	97.50	95.20	26.04	4.23
BFRNet(ours)	**98.75**	**95.20**	**7.68**	**1.90**

TABLE II
EXPERIMENTS ON THE 30-CLASSIFIED AND 24-CLASSIFIED AID DATASET

Number of datasets containing classes	AID Top-1(%)
AID 30 Classes 10000	**95.20**
AID 24 Classes 8060	96.54

extraction. Although more global information is obtained by window sliding, the classification effect is not satisfactory. CoAtNet adopted a CNN-Transformer hybrid architecture for image classification, whose classification accuracy on two datasets is about 2% lower than our method, but its parameters and GFLOPs are much higher. For the traditional CNN classical network model, its accuracy in the image classification task is slightly lower than recent ViTs. Resnet50 is weaker than BFRNet, and its training cost is higher. In contrast, BFRNet shows excellent performance, and it incorporates a channel attention mechanism in ViTs, as well as the use of convolution. It captures both local detail information and establishes global dependencies, and it significantly outperforms previous work with a lower parameters and computational complexity. In addition, as shown in Table I, the classification accuracy of AID dataset differs more than that of RSSCN7. Particularly, there are lower precision of five classes on AID dataset with particularly dense image scenes (as shown in Fig. 9). We performed the experiment again after removing the five classes, and the classification accuracy was significantly higher, as shown in Table II. We think that this is due to the huge difference between the two datasets, resulting in a significant difference in classification accuracy.

The confusion matrix provides a clear view of the classification of each category for a visual analysis of the classification error. Fig. 10 shows the confusion matrix generated by BFRNet on the RSSCN7 dataset. It can be seen that classes in the dataset are basically concentrated on the diagonal lines, indicating that these images are correctly

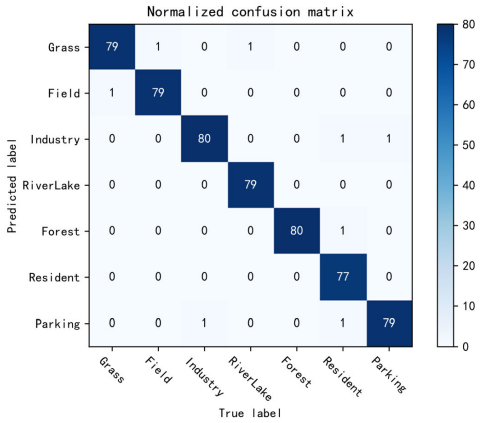


Fig. 10. Confusion matrix of BFRNet on the RSCNN7 dataset.

classified into the classes to which they belong. The overall error rate of BFRNet for the RSSCN7 validation set is only 1.25%, indicating that BFRNet can effectively classify the RS images. The AID dataset includes 30 classes, and the confusion matrix is shown in Fig. 11.

C. Ablation Experiments and Visualization

In this section, we describe the ablation experiments as a way to show that the key components of BFRNet are effective.

First, we conduct an experiment with the BFR block stacking layer settings for each of the four stages included in the network. Considering the reference validity of the classification performance, stages 1–4 [1, 2, 4, 1] are the baseline, and we gradually increase the number of layers in each stage, so as to obtain the layer settings with the highest classification accuracy. As shown in Table III, the highest classification accuracy is obtained for the superimposed layer settings of stages 1–4 [2, 2, 6, 2]. Its parameters and computation are moderate, and classification effect is poor when the number of layers increases further. The experiment illustrates that our network settings are optimal.

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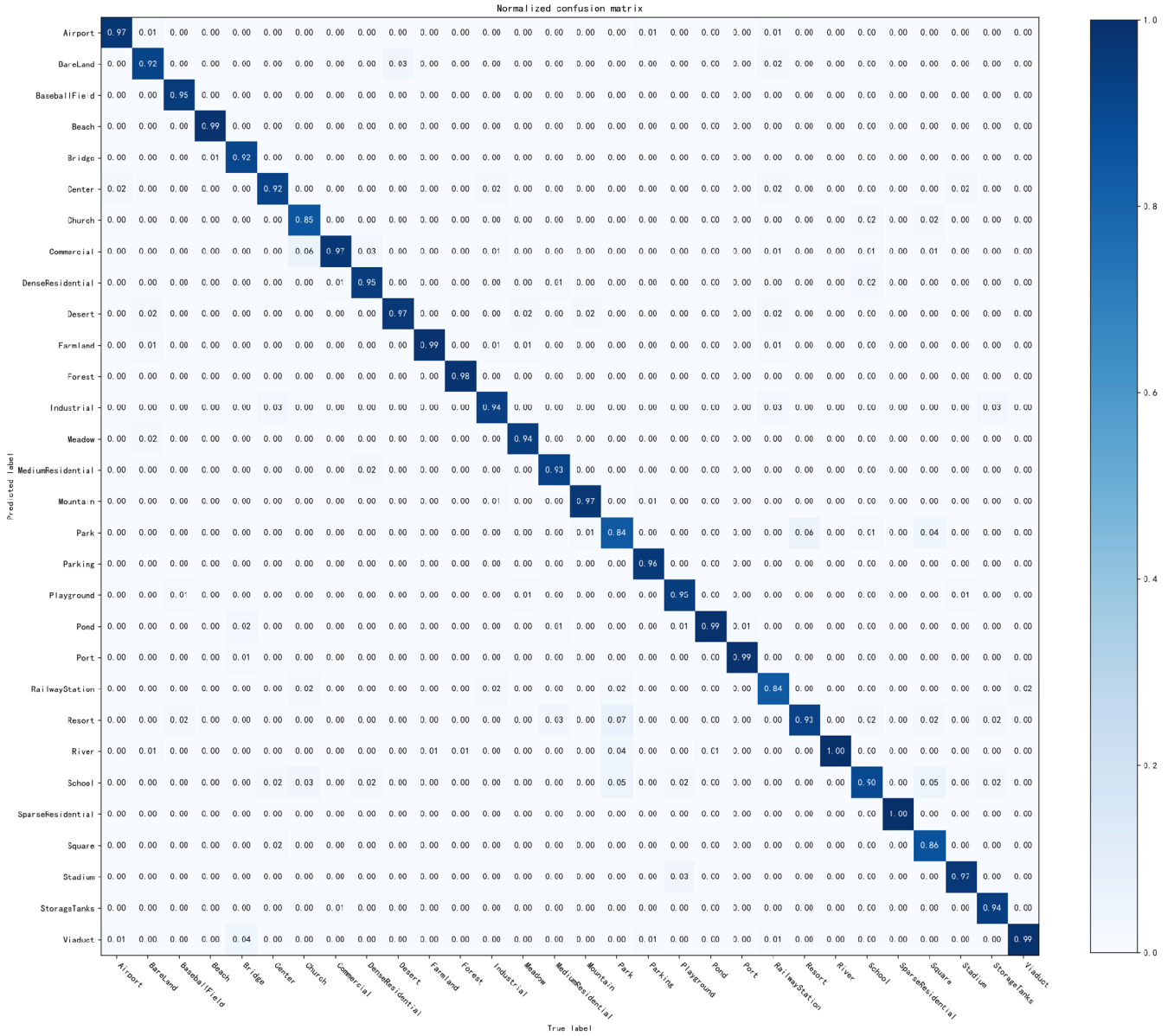


Fig. 11. Confusion matrix of BFRNet on the AID dataset.

TABLE III
MODEL NUMBER OF STACKED LAYERS OF BFR BLOCK IN EACH OF THE FOUR STAGES

Number of layers setting Stage1~4	Top-1 (%)	Params (M)	FLOPs (G)
[1,2,4,1]	97.50	4.85	1.29
[1,2,5,2]	98.39	7.07	1.44
[2,2,6,2]	**98.75**	**7.68**	**1.90**
[2,2,8,3]	98.21	12.76	2.55
[2,3,8,4]	97.68	26.58	4.73

Table IV shows the ablation experiment of different module superpositions. First, only the pyramid pool self-attention branch is retained, and its classification accuracy is 93.75%. Then, the local window attention branch is added, and it can be seen that its accuracy increases significantly. After that, the linear attention module is added for feature fusion, and its accuracy increases further. Finally, by adding the channel attention used in combination with the local window attention module, the accuracy increases by about 2%. For FFN part, after the BGU module used to replace the traditional MLP, the model structure is complete and reaches the highest accuracy of 98.75%. The above sets of data illustrate that each module of BFRNet works effectively.

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TABLE IV
ABLATION EXPERIMENT OF DIFFERENT MODULE SUPERPOSITIONS

Attention				FFN	Top-1	Params	FLOPs
Channel Attn(a)	Local-win Attn(b)	Global Attn(c)	Linear Attn(d)	BGU (e)	(%)	(M)	(G)
×	×	√	×	×	93.75	5.35	1.40
×	√	√	×	×	95.00	5.55	1.46
×	√	√	√	×	95.89	6.15	1.65
√	√	√	√	×	97.50	7.38	1.67
√	√	√	√	√	**98.75**	**7.68**	**1.90**

TABLE V
ABLATION EXPERIMENTS OF MODULES IN BFR BLOCK. MODULES (a) IS CHANNEL ATTENTION, (b) IS LOCAL WINDOW ATTENTION, (c) IS PYRAMID POOL SELF-ATTENTION, (d) IS LINEAR ATTENTION, AND (e) IS BIDIMENSIONAL GATING UNIT

Attention				FFN	Top-1	Params	FLOPs
Channel Attn(a)	Local-win Attn(b)	Global attn(c)	Linear Attn(d)	BGU (e)	(%)	(M)	(G)
√	√	√	×	√	96.79	7.46	1.71
√	√	√	√	×	97.50	7.38	1.67
×	√	√	×	√	94.46	5.94	1.69
×	√	√	√	×	95.54	5.87	1.65
√	√	×	×	√	94.29	6.73	1.58
√	√	√	×	×	96.96	6.47	1.48
×	√	×	×	√	95.36	5.56	1.56
√	√	×	×	×	93.04	6.08	1.35
√	√	√	√	√	**98.75**	**7.68**	**1.90**

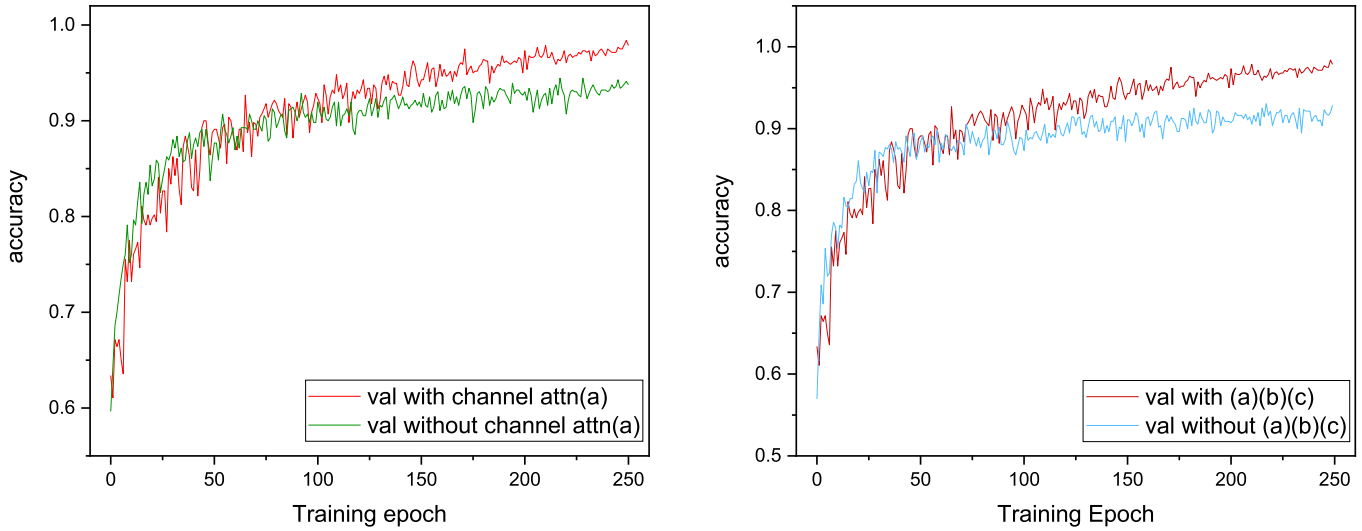


Fig. 12. Left: line graphs for the discarded module (a) and with (a); right: line graphs for the accuracy of removing the module (a), the window attention module (b), the linear attention module (c), and with these three modules.

In Table V, the validity of the modules is verified by removing one or two or three modules each time. Eight scenarios are described in Table V. The original two modules are removed in three to six rows, and the classification accuracy decreases by 4.29% when the channel attention modules (a) and (d) are discarded. This result indicates that it is necessary for us to extract features from both the spatial dimension and the channel dimension in the model design. Removing the original three modules of the model in seven and eight rows also reduce classification accuracy. Among them, when modules (c)–(e) are removed, the accuracy decreases by 5.71%, which indicates that pyramid pool self-attention is effective in extracting global features. In addition, multiple sets of experiments demonstrate the necessity of the existence of linear attention with BGUs.

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TABLE VI

ABLATION EXPERIMENTS OF VIT BLOCK AND BFR BLOCK. VIT BLOCK CONSISTS OF MHSA AND MLP. BFR BLOCK CONSISTS OF BFR-MHSA AND BGU

ViT		BFRNet		Top1	Params	FLOPs
MHSA	MLP	BFR-MHSA	BGU	(%)	(M)	(G)
√	√	×	×	94.29	4.34	1.24
√	×	×	√	95.00	4.98	1.48
×	√	√	×	97.68	7.04	1.66
×	×	√	√	**98.75**	**7.68**	**1.90**

TABLE VII

CROSS-VALIDATION EXPERIMENTS ON THREE DATASETS TO DISCUSS THE TRANSFERABILITY OF BFRNET WITH PARTIAL DOMAIN ADAPTATION METHODS. DATASETS A: AID, U: UCMERCED, AND N: NWPU-RESISC45

Method	N->A(%)	N->U(%)	A ->U(%)	A->N(%)
PDA [60]	89.54	80.67	64.62	84.61
BFRNet	**90.05**	**95.24**	**87.14**	78.05

Through Table V, it can be seen that the channel attention module (a) is crucial to improve model performance. BFRNet is a network based on a BFR, where the information complement of the channel dimension is very important. In Fig. 12 left, we give the line graphs for the discarded module (a) and with (a). Fig. 12 right shows the line graphs for the accuracy of removing the module (a), the window attention module (b), the linear attention module (d), and with three modules. As can be seen in Fig. 12, the model accuracy is significantly reduced after removing the correlation module used for capturing channel information.

In Table VI, we perform ablation experiments on ViT block with our proposed BFR block. Among them, MHSA and MLP are the main components of ViT block, while BFR-MHSA and BGU are the main components of BFR block. As shown in Table VI, the accuracy of the ViT block is lower than BFR block after the four stages, and the accuracy increases when the MLP block is replaced by the BGU in the BFR block. When the BFR-MHSA in the BFR block is combined with the MLP block of the ViT block, the accuracy improved by 3.39% compared to the initial ViT block. The results show that our BFRNet is more efficient compared to ViT.

In recent years, domain adaptive methods have been proposed to address cross-region, multitemporal, and multisensor RS applications, and many works make great results in domain adaptive RS image classification. It is important in practical applications and more focused on testing the transferability of the model. In order to verify the transferability and generalization of our method on different datasets, we compare it with the latest method, called partial domain adaptation [60]. From the experimental results, as shown in Table VII, our method is better than the comparison method on three scenarios and worse than the comparison method on one indicator. Although the results may vary due to differences in datasets, such as different classes or different overall features of the same class of images in different datasets, it also indicates to some extent that our method has good transferability.

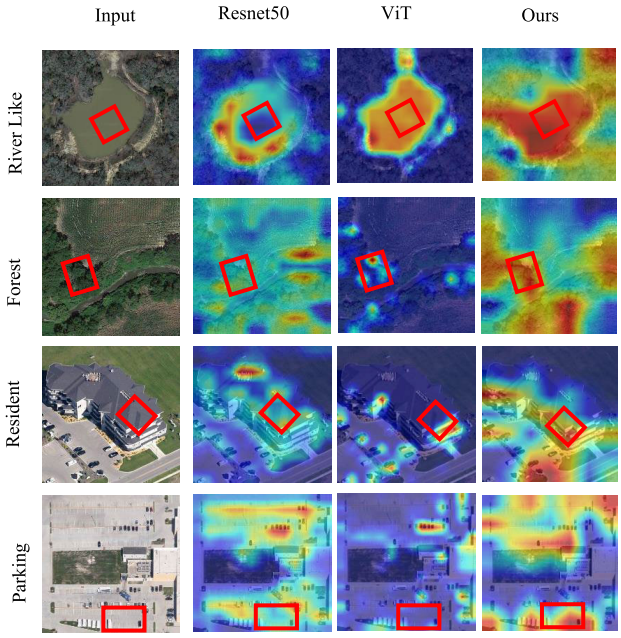


Fig. 13. Heat map plotted on RSSCN7 dataset, compared with Resnet50 and ViT.

In order to verify the effectiveness of BFRNet more intuitively, the heat map of our method versus other methods is given in Fig. 13, and the activated heat map is generated by Grad-CAM [54]. The heat map represents the distribution of the contribution of the predicted output of the neural network, and the darker red color indicates the higher value, the higher response, and contribution to the network. From Fig. 13, it can be seen that our method has a darker red color in the target region compared with resnet50 and ViT, which means that this region is more important for the classification in the output prediction. However, the corresponding red region of Resnet50 in Fig. 13 does not show distinctly, which indicates that its prediction output has a low score and poor performance in this classification. For ViT, its red region does not accurately

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identify the target region of this class and has a lighter red, which indicates that its prediction output has a lower score too. Therefore, it can be seen from Fig. 13 that our method is better than ViT and ResNet50. In addition, the selected boxes in Fig. 13 are some areas that are particularly important for classification, and the detailed information obtained by different methods can be seen in the heat map. For the class "parking," both ViT and Resnet50 do not recognize the parking space in the box, while our method is darker red here. This is also true for other classes, which shows that our method is more accurate for classes identification.

V. Conclusion

BFRNet is proposed to harvest the powerfulness of deep neural network models for RS image classification. It is a CNN-Transformer hybrid architecture, which first uses a BLWS to extract local spatial and channel information. Then, the global information is extracted by using pyramid pool self-attention. Finally, global–local information is fused by the intersection of linear attention modules. The modules are formed into a complete BFR block, and then, this block is superimposed in four stages to finally output the classification results. Experiments are conducted on RSSCN7 and AID RS image datasets, and the results show that the accuracy of BFRNet reaches 98.75% and 95.20%, respectively. Compared with other methods of image classification, BFRNet can effectively extract local feature and long-range global dependencies relationship while reducing the parameters and computational cost. It also obtains higher image classification accuracy, thus verifying the efficiency of BFRNet. CNN is a classic method, and ViT is a recent hot topic in machine learning and computer vision; various improvements have been made over the past few years, and they are used in combination to exploit the potential. In particular, the idea of joint attention embodied in the model can be flexibly used in the existing methods to improve the performance of deep neural network, which deserves to investigate as a possible future work.

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